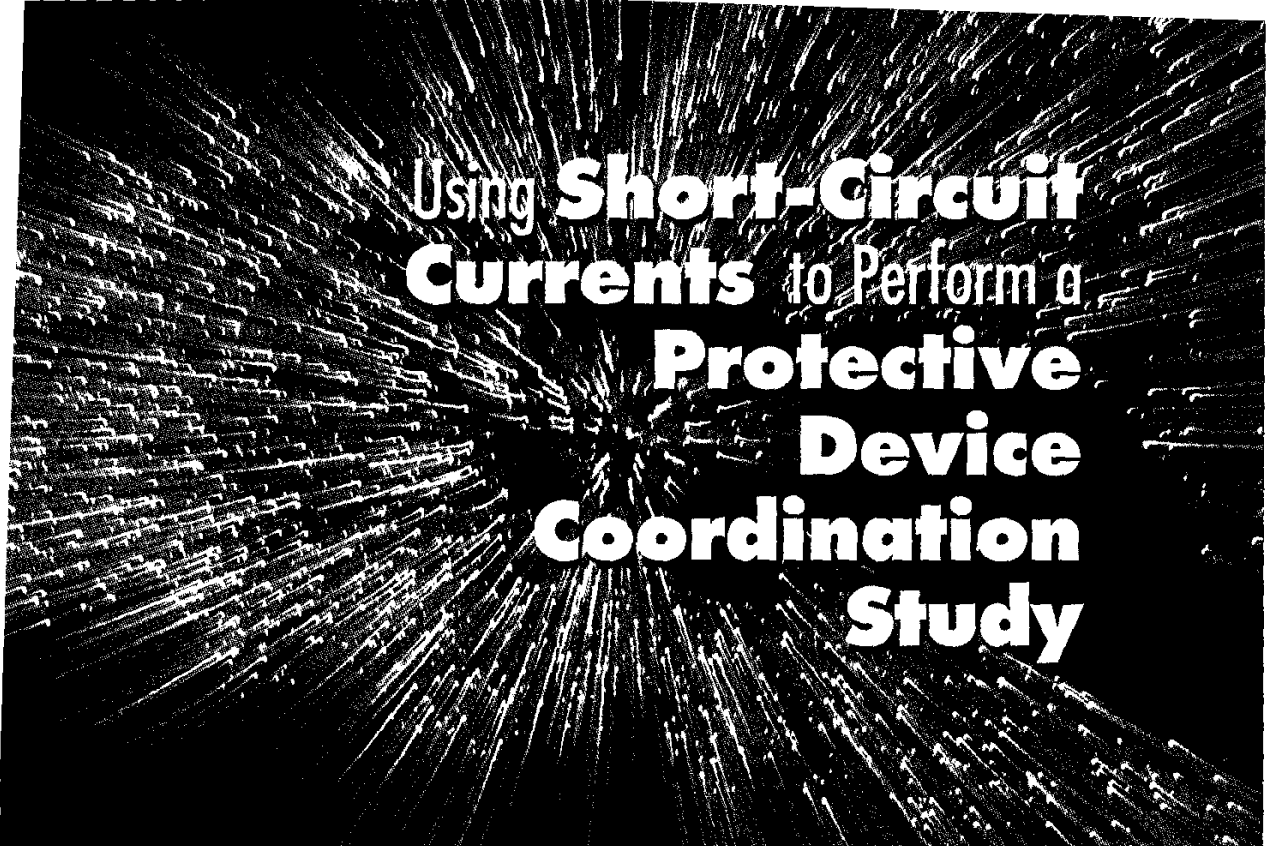




Using Short-Circuit Currents to Perform a Protective Device Coordination Study

There are two types of short-circuit studies of interest to the power engineer. The first determines the first-cycle (momentary) and contact-parting (interrupting) short-circuit current duties (i.e., asymmetrical rms or peak currents) at the buses of the power system, which are used to select the short-circuit withstand and interrupting capabilities of switchgear [1, Chap. 4]. To ensure the safety of personnel and to prevent damage to power system apparatus, this type of short-circuit study is performed under worst-case assumptions. These assumptions include selecting the worst-case fault location at the switchgear bus (which receives all possible short-circuit current contributions) and the fault type (usually a bolted, three-phase fault) [1, Sec. 4.3.2].

The second type of study determines the subtransient and transient short-circuit currents that an overcurrent-protective device will sense in order to initiate the prompt removal of the affected portion of the power system by its circuit interrupter [2, Sec. 1.5]. These short-circuit currents are necessary to properly select the instantaneous

and time-delay settings of the overcurrent-protective devices [1, Chap. 5], [2, Chap. 14], [3, Chap. 9], [5, Chap. 9].

Oftentimes, the engineer possesses commercial software for the first type of short-circuit study. To perform the computations, many of the algorithms employ either reduction of the complex impedance network representation of the power system [2, Sec.

2.6], or else reductions of separate reactance and resistance networks [1, Chap. 4]. Thus, a natural question is whether the output from such a computer-aided short-circuit study can also be applied to the second type of study. The pur-

pose of this article is to address this question by comparing the two types of short-circuit studies. The basis for the comparison is the complex impedance network method.

The organization of the article is as follows. The next section provides a brief background on the fundamentals of short-circuit analysis. Following that background is a discussion of how various types of phase overcurrent protective devices respond to the short-circuit current waveform. Based on this discussion, the next section specifies which

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The risk of injury to personnel and damage to power system apparatus is high if a short-circuit study is performed by someone who is unqualified or inexperienced.

short-circuit currents are necessary to perform a protective device coordination study. A comparison of the two types of short-circuit studies follows, based on the complex impedance network representation of the power system. The results of the comparison are then used to present recommendations for how the results of a computer-aided study to compute short-circuit current duties can be applied to determine the short-circuit currents necessary to select the settings of phase overcurrent protective devices. Finally, a sample industrial power system is provided to clarify the recommendations.

A few words of caution are in order before we proceed. The risk of injury to personnel and damage to power system apparatus is high if a short-circuit study is performed by someone who is unqualified or inexperienced. Simply reading the literature on the subject does not certify or qualify an engineer to perform power system studies in practice. Although commercial software is the quickest and most efficient means of performing a short-circuit study, it is important that the engineer know how to perform these calculations by hand, in conjunction with using the commercial software. The results of a computer-aided short-circuit study are susceptible to errors. The most common ones are human errors (e.g., data-input errors, misinterpretation of results, etc.); although, occasionally the software is the culprit. Nevertheless, the engineer, not the software vendor, is responsible for the accuracy of the study.

Background

This section provides only a brief background on the fundamentals of short-circuit analysis to serve the purpose of this article. References [1]-[5] provide further details.

Fig. 1 illustrates a typical short-circuit current waveform for one phase of a three-phase synchronous generator, previously unloaded, which has been subjected to a balanced, three-phase fault across its accessible terminals. In general, an asymmetrical, short-circuit current waveform is produced by a balanced, three-phase fault.

Asymmetrical alternating currents when treated as a single current wave are difficult to interpret for circuit breaker applications and relay-setting purposes. Complicated formulas are required to calculate their magnitude unless resolved into components. The asymmetrical alternating currents are, for circuit breaker applications and re-

lay-setting purposes, arbitrarily divided into simple components, which makes it easy to calculate the short-circuit magnitude at certain significant times after the short-circuit occurs. [5, pp. 15 and 16]

For this reason, the asymmetrical short-circuit current waveform of Fig. 1 has been decomposed into two components: a unidirectional dc component and a symmetrical ac component. The dc component exponentially decays to zero, and the amplitude of the symmetrical ac component eventually decays to a constant amplitude in the steady-state. The sum of these two components at any time instant is equal to the total asymmetrical short-circuit current at that instant. Furthermore, it is important to note that the asymmetrical short-circuit current waveform of the other phases, which are not shown in Fig. 1, differ only in their dc components. The dc components have different initial values, but decay exponentially at the same rate. The rate of exponential decay (i.e., time constant) of the dc component is related to the short-circuit X_{eq}/R_{tq} ratio, where X_{eq} and R_{tq} are the equivalent reactance and resistance at the fault location, respectively. The envelope of the symmetrical ac component is identical for all three phases.

To simplify the analysis, steady-state ac circuit theory is exploited to calculate a worst-case (i.e., initial) root-mean-square (rms) magnitude of the symmetrical ac component to characterize a particular time interval of the fault-on time period. There are three time intervals of interest:

... the subtransient period, lasting only for the first few cycles (at 60 Hz), during which the current decrement (i.e., decay of the asymmetrical short-circuit current waveform) is very rapid; the transient period, covering a relatively longer time (i.e., beyond 6 cycles), during which the current decrement is more moderate; and finally the steady-state period (i.e., beyond 30 cycles) ... [6, Sec. 6-6]

To simplify verbiage, the expression "worst-case (i.e., initial) rms magnitude of the symmetri-

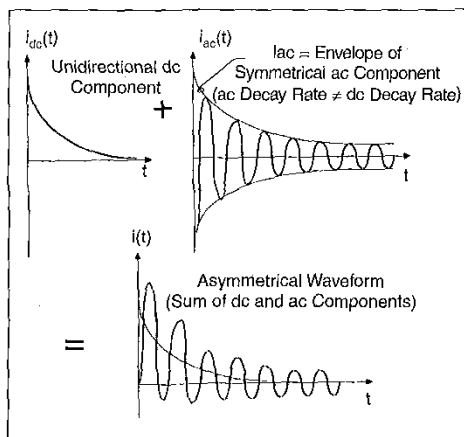


Fig. 1. Phase current of a short-circuited generator.

cal ac component" is hereafter shortened to "symmetrical rms current."

Thus, the purpose of the analysis is to calculate the following: I_d'' = subtransient symmetrical rms current, I_d' = transient symmetrical rms current, and I_d = steady-state symmetrical rms current. In order to perform these calculations, a constant reactance for each time period is assigned to every machine: X_d'' = direct-axis subtransient reactance, X_d' = direct-axis transient reactance, and X_d = direct-axis synchronous reactance.

On the other hand, the utility generators are usually remote from the plant. Their short-circuit current contributions to the remote plant appear to them as a small increase in load current. Furthermore, these current contributions tend to remain constant during the entire fault-on time period. Therefore, the utility tie is often modeled by a constant (i.e., Thevenin equivalent) impedance over the entire fault-on time period, which is determined by knowing the available short-circuit contribution that can be delivered to the plant. Finally, the passive apparatus of the system (i.e., transformers, cables, busway, reactors, etc.) are modeled by impedances that are constant over the entire fault-on time period.

The problem of calculating symmetrical rms current can be summarized as follows:

- 1. Construct an impedance network by interconnecting the appropriate impedances of the apparatus in accordance with the one-line diagram.
- 2. Impose the fault location on the network.
- 3. Use the series, parallel, and delta-wye transformation formulas to reduce the network to an equivalent impedance, Z_{FQ} .
- 4. Use Ohm's Law to divide $|Z_{FQ}|$ into the rms magnitude of the prefault phase-to-neutral voltage at the faulted bus to yield the symmetrical rms current.

In order to apply the above procedure to calculate subtransient symmetrical rms current, use direct-axis subtransient reactances for all of the rotating machines in the subtransient network. To apply the procedure to calculate transient symmetrical rms current for several cycles (i.e., beyond six cycles) after fault inception, the synchronous machine (i.e., generator and motor) model uses either its transient reactance or a larger impedance related to the magnitude of decaying short-circuit current at the specified calculation time [1, Sec. 4.5.4.3]; all induction motor contributions are omitted in the transient network.

At this point, we have only accounted for the ac component of the asymmetrical short-circuit current waveform.

Calculation of the precise rms value of an asymmetrical current at any time after the inception of a short-circuit may be very involved. Accurate decrement factors to account for the dc component at any time are required, as well as factors for the rate of change of the apparent

reactances of the generators. This precise method may be used, if desired, but simplified methods have evolved whereby the dc component is accounted for by simple multiplying factors. The multiplying factor converts symmetrical rms current into asymmetrical rms current, which accounts for both the dc and ac components. [5, p. 20]

The short-circuit currents necessary to select the settings of an overcurrent-protective device must reflect the operating characteristics and times of the device. The response of the protective device to subtransient asymmetrical rms current and/or subtransient asymmetrical peak current must be known. On the other hand, it is unnecessary to account for the negligible dc component over the transient period; only the transient symmetrical rms current is necessary.

The subtransient asymmetrical rms current is calculated by multiplying the subtransient symmetrical rms current by an asymmetry multiplying factor. Strictly speaking, the asymmetry multiplying factor depends on the short-circuit X_{FQ}/R_{FQ} ratio of the Thevenin equivalent impedance, Z_{FQ} , at the location [4], [7], [9]. In practice, a multiplying factor of 1.6 is used on medium-voltage systems (>1000 V) and 1.5 on low-voltage systems [5, p. 548]. The exception for medium-voltage systems is to use the smaller multiplying factor of 1.5 on systems of 2.4 kV to 5 kV when there are transformers or there is appreciable line resistance between the fault point and the source [8, Section .54, p. 2]. It is important to note that these figures may need to be increased to avoid nuisance tripping.

Protective Device Operation

The operating characteristic of a phase-overcurrent-protective device dictates how it responds to the asymmetrical short-circuit current waveform. For a static relay, it is important to know whether it is programmed to sense rms, average, or peak current, and is calibrated for rms current.

Induction-type relays are affected only by the ac component of the short-circuit current waveform. Generally, the time-current characteristic curve of such a device is not plotted beyond the subtransient symmetrical rms current that it will experience.

There are other instantaneous phase-overcurrent-protective devices that are affected by both the dc and ac components of the short-circuit current waveform, and respond within the first half-cycle. Consequently, subtransient asymmetrical rms current is a crucial factor in setting these devices.

Finally, the transient symmetrical rms current is used to verify the coordination of time-delay phase-overcurrent-protective devices that are set to operate slower than 0.1 s (i.e., six cycles).

The following list has been adapted from Sec. 4.5.4.3 of [1]; pp. 548-9 of [5]; and Table 2, Section .541 of [8]. It summarizes which short-circuit currents affect the operation of specific types of phase-overcurrent-protective devices.

Subtransient Symmetrical RMS Current

These include high-speed, induction-type relays that operate in less than three cycles (0.05 s); short-time induction relays that are set to operate faster than 0.1 s (six cycles); generator, transformer, and bus differential relays; power-line current balance relays; and pilot-wire relays.

Subtransient Asymmetrical RMS Current

Instantaneous (i.e., magnetic) functions of thermal-magnetic-molded or insulated-case circuit breakers; plunger or hinged armature-type instantaneous relays; fast-melting-time zone (<0.1 s) of fuses; and instantaneous and short-time delay functions that are set to operate faster than 0.1 s on electromagnetic or static (i.e., solid-state or microprocessor-based) direct-acting trips of low- and medium-voltage circuit breakers are examples of this current.

Transient Symmetrical RMS Current

These include time-delay induction-type relays that are set to operate slower than 0.1 s; fuse operation, under sustained fault current, in melting-time portion of the characteristic that is slower than 0.1 s; and long-time and short-time delay functions that are set to operate slower than 0.1 s on electromagnetic or static (i.e., solid-state or microprocessor-based) direct-acting trips of low- and medium-voltage circuit breakers.

Short-Circuit Currents To Select Settings

Based on the previous discussion, the subtransient symmetrical rms short-circuit current under the maximal power system configuration should be calculated at the appropriate fault location of every phase-overcurrent-protective device. The maximal power system configuration for the subtransient network includes all possible short-circuit current sources (i.e., utility service supply, in-plant generation, and motors). Generally, the time-current characteristic curve of an induction-type relay is not plotted beyond the maximal subtransient symmetrical rms current nor the pickup setting of an instantaneous phase-overcurrent-relay in the same circuit, whichever is smaller.

Knowledge of the maximal subtransient asymmetrical rms current is necessary to determine the settings of instantaneous-phase-overcurrent-protective devices that are responsive to both the dc and ac components of the short-circuit current waveform. Minimum settings are required to allow motor locked-rotor current or transformer magnetic inrush current. Higher settings are not necessary, but can be used. Coordination of instantaneous-phase-overcurrent devices in series at the same voltage level is not possible unless there is significant impedance between buses, caused by long cable runs or current-limiting reactors. In this case, the pickup setting of the upstream device has to be

about twice the pickup setting of the downstream device. (Note: In this context, "pickup" refers to the effective primary pickup that has taken the current transformer ratio into account [10].) A more accurate criterion is that the subtransient asymmetrical rms current for a fault at the downstream device must be less than the instantaneous setting of the upstream protective device.

If the subtransient asymmetrical rms current under the minimal power system configuration is less than the pickup setting of an instantaneous overcurrent relay, the relay will not operate, and protection will depend upon time-delay phase-overcurrent relays. This situation requires the calculation of minimal subtransient symmetrical rms current in order to determine if the time-delay overcurrent relays will provide adequate protection. The minimal power system configuration for the subtransient network may apply at night or when production lines are down for any reason. Note that this network still uses direct-axis subtransient reactances for all of the rotating machines in use.

The transient symmetrical rms short-circuit current under the maximal power system configuration is calculated at the appropriate fault location of every time-delay phase-overcurrent-protective device that is set to operate slower than 0.1 s (six cycles). The maximal power system configuration for the transient network neglects induction motor contributions, but includes all other short-circuit current sources (i.e., utility service supply, in-plant generation, and synchronous motors). Knowledge of the maximal transient symmetrical rms current is necessary to establish the coordination time intervals for electromagnetic (i.e., induction-disc) or static (i.e., solid-state or microprocessor-based) time-delay phase-overcurrent relays with other phase-overcurrent-protective devices.

Finally, it may be important to calculate the transient symmetrical rms current under the minimal power system configuration (i.e., minimal transient symmetrical rms current) in order to investigate whether sufficient fault current is available to cause operation of a time-delay overcurrent-protective device within a satisfactory time. The minimal power system configuration for the transient network may apply at night or when production lines are down for any reason, and it is often defined under minimum power supply. Note that the transient network under the minimal power system configuration still neglects induction motor contributions, but includes all other short-circuit current sources (i.e., utility service supply, in-plant generation, and synchronous motors).

The following list summarizes the short-circuit currents that should be calculated in order to perform a coordination study:

Maximal and Minimal Subtransient Symmetrical RMS Currents: High-speed, phase-overcurrent-protective

devices that are responsive only to the ac component of the short-circuit current.

Maximal Subtransient Asymmetrical and Minimal Subtransient Symmetrical RMS Currents: Instantaneous phase-overcurrent-protective devices that are responsive to both the dc and ac components of the short-circuit current.

Maximal Subtransient and Maximal and Minimal Transient Symmetrical RMS Currents: Time-delay phase-overcurrent-protective devices that are responsive to both the dc and ac components of the short-circuit current, and are set to operate slower than six cycles at 60 Hz.

Comparison of Studies

Let us first consider the short-circuit study whose results are used to select the short-circuit withstand and interrupting capabilities of medium-voltage and low-voltage circuit breakers and fuses from manufacturers' tables. For this application, there are two types of networks to calculate symmetrical rms current: first-cycle (momentary) and contact-parting (interrupting).

The momentary network is identical to the subtransient network, and it is used to calculate the symmetrical rms current over the subtransient time interval of the fault-on time period. An appropriate asymmetry multiplying factor is then used to convert momentary symmetrical rms current into momentary asymmetrical rms current. The withstand capability of a medium-voltage circuit breaker, the interrupting capability of a fuse, and the interrupting capability of a low-voltage circuit breaker must exceed the momentary asymmetrical rms current [1, Chap. 4].

The interrupting network is used to calculate the symmetrical rms current for medium-voltage circuit breaker minimum contact-parting times of 1.5 to 4 cycles after the inception of the short-circuit. For this network, direct-axis subtransient reactances are used for the in-plant synchronous turbine-generators, synchronous motor reactance is calculated by multiplying its direct-axis subtransient reactance by a rotating machine reactance multiplier of 1.5, and induction motor contributions are included in accordance with Tables 4-1 and 4-2 of the *IEEE Red Book* [1]. An appropriate asymmetry multiplying factor is then used to convert interrupting symmetrical rms current into interrupting asymmetrical rms current. The interrupting capability of a medium-voltage circuit breaker must exceed the interrupting asymmetrical rms current.

Let us now compare the two types of short-circuit studies. The basis for the comparison is the complex impedance network representation of the power system.

Although the momentary network is identical to the subtransient network, fault location is the principle reason why the momentary symmetrical

rms current is not the same value as the subtransient symmetrical rms current.

In order to select the short-circuit withstand and interrupting capabilities of circuit breakers and fuses at a switchgear bus, a conservative practice is to assume the maximal power system configuration and to locate the three-phase short-circuit at the switchgear bus so as to receive all possible short-circuit current contributions. The maximal power system configuration for the first-cycle network includes all possible short-circuit current sources (i.e., utility service supply, in-plant generation, and motors).

On the other hand, to select the time-current characteristics and settings of a phase-overcurrent-protective device, the three-phase short-circuit is located at the terminal of the device (i.e., or its current transformers), which causes the largest short-circuit current to flow through it. (For a radial power system, this fault location is the load-side terminal of the protective device or its current transformers.)

Let us consider a specific example to better illustrate the distinction between these two types of fault locations. Fig. 2 presents the designations and illustrates the directions of the short-circuit current contributions to the three-phase fault location, F_3 , at the 4.16 kV substation 3 bus. The purpose of this fault location is to calculate the momentary symmetrical rms current. In Fig. 2, I denotes the phasor representation of the short-circuit current contribution; the superscript, f , refers to the first-cycle network. Finally, $I_{F_3}^f = |I_{F_3}^f|$ is multiplied by an appropriate asymmetry multiplying factor [1, Chap. 4], based on the equivalent short-circuit X/R ratio at the fault location, to yield the first-cycle (momentary) asymmetrical rms current

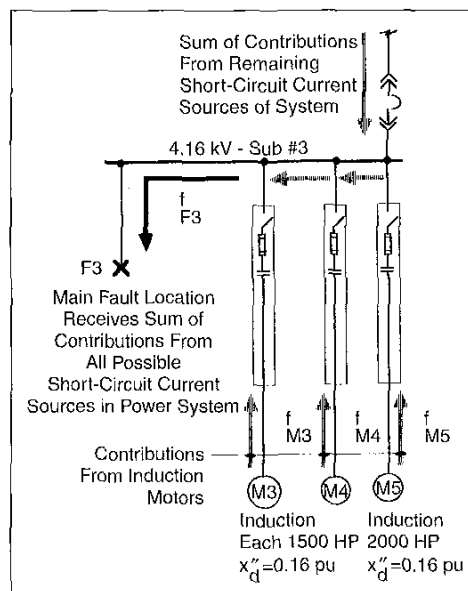


Fig. 2. Three-phase fault location to select interrupting capabilities of power fuses.

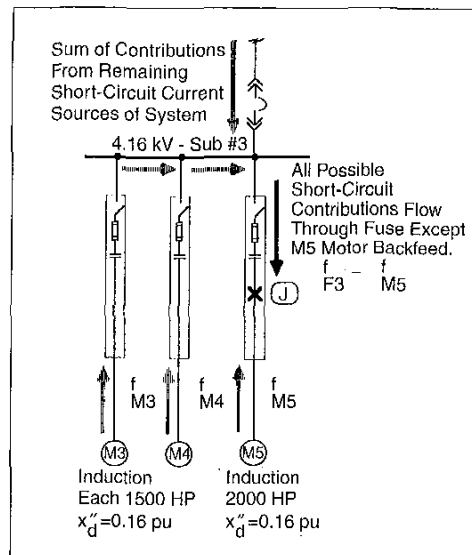


Fig. 3. Fault location to select time-current characteristic of fuse.

(i.e., momentary duty). The short-circuit interrupting capabilities of the medium-voltage power fuses at this bus must exceed the momentary duty at this bus.

Next, Fig. 3 shows the three-phase fault location, J, at the load-side terminal of the power fuse for induction motor M5. The purpose of this location is to calculate the subtransient symmetrical rms current. Using an appropriate asymmetry multiplying factor (cf. "Background" section), the subtransient asymmetrical rms current delineates the upper (i.e., right-most) current limit of the fuse's time-current characteristic curve.

Now, the first-cycle network is predominantly reactive because the short-circuit X/R ratios of its apparatus are relatively large. Thus, as an approximation, the magnitudes of the phasor currents can be added and subtracted without regard to their phase angles (which are all in the neighborhood of 90°). Therefore, referring to Fig. 3, the subtransient symmetrical rms current at fault location J can be approximated by: $I_d'' \cong I_{F3}^f - I_{M5}^f$. Finally, as stated previously, $(I_d'')_f$ is multiplied by an appropriate asymmetry multiplying factor (cf. "Background" section) to yield the subtransient asymmetrical rms current.

Note that the difference between the subtransient symmetrical rms current and the momentary symmetrical rms current is the short-circuit current contribution (i.e., first-cycle symmetrical rms current) of induction motor M5 (i.e., motor backfeed). For this example, the motor backfeed is approximated by:

$$\begin{aligned}
 I_{M5}^f &= (I_{M5}^f)_{\text{per-unit}} \times (\text{base amps})_{M5} \\
 &\cong \frac{1.0}{0.16} \frac{2000}{\sqrt{3}(4.0)} = 1804 \text{ A @ 4.0 kV.}
 \end{aligned}$$

Finally, the principle difference between the contact-parting (interrupting) network and the transient network is time. Medium-voltage circuit breaker minimum contact-parting times are 1.5 to 4 cycles after the inception of the short-circuit. Induction motor contributions are included in the contact-parting (interrupting) network. The transient period covers a relatively longer time period, beyond six cycles, so that, induction motor contributions are ignored in the transient network. Therefore, the interrupting symmetrical rms current is larger by induction motor contributions than the transient symmetrical rms current.

Recommendations

In this section, we present recommendations on how to apply the results of a computer-aided short-circuit study that selects withstand and interrupting capabilities of switchgear to determine the short-circuit currents (i.e., maximal and minimal subtransient and transient symmetrical rms currents) to properly select the time-current characteristics and settings of phase-overcurrent-protective devices. It is very important to note that these recommendations are based on the complex impedance network representation of the power system.

First and foremost, it is important to use the appropriate fault location and to account for motor backfeed. Imposing the appropriate fault location and accounting for motor backfeed on the subtransient and first-cycle networks, the subtransient symmetrical rms current is equal to the momentary symmetrical rms current, which can be calculated by the procedure in Chapter 4 of the *IEEE Red Book* [1]. In turn, the subtransient asymmetrical rms current is calculated by multiplying the subtransient symmetrical rms current by an appropriate asymmetry multiplying factor, in accordance with the rules discussed above (cf. "Background" section).

As mentioned previously, the transient symmetrical rms current is different (i.e., smaller by induction motor contributions) than the interrupting symmetrical rms current. However, the principal use of the maximal transient symmetrical rms current is to establish the coordination time interval for a time-delay phase-overcurrent relay (i.e., set beyond six cycles) with another phase-overcurrent-protective device. Establishing this time interval at a larger value of short-circuit current is a conservative practice insofar as coordination is concerned. However, a coordination time interval that is too conservative may not be desirable from the standpoint of limiting equipment damage. In the event that the downstream protective device fails to operate, the upstream (backup) protective device may take too long to operate. In summary, with regard to coordination, it is permissible [2, Sec. 14.2.2] to use the maximal interrupting symmetrical rms current (instead of the maximal transient symmetrical rms current) to establish the coordination time interval.

Finally, it is permissible [1, Secs. 4.5.4.3 and 4.6.5] to use the approximate 30-cycle network (instead of the minimal power system configuration for the transient network) to investigate whether the minimum symmetrical rms current is sufficient to cause satisfactory operation of a time-delay current-actuated relay. The approximate 30-cycle network is a minimum source representation that may apply at night or when production lines are down for any reason. Some of the source circuit breakers may be open, and all motors are omitted. In-plant generators use direct-axis transient reactance or a larger reactance related to the magnitude of decaying generator short-circuit current at the specified calculation time. The dc component will have decayed to near zero and is not considered.

Example

The preceding recommendations are applied to the sample industrial power system of Fig. 4. Fig. 4 shows the appropriate fault locations (i.e., labeled A through K) of the phase-overcurrent-protective devices, and provides the input data and the labeling of buses for the computer-aided short-circuit study. The commercial software is designed to compute the momentary and interrupting short-circuit current duties (i.e., asymmetrical rms currents) at the buses of the power system, which are used to select the short-circuit withstand and interrupting capabilities of switchgear. The algorithm of the software employs complex impedance network reduction.

The program was run under three different power system configurations: maximal, minimal,

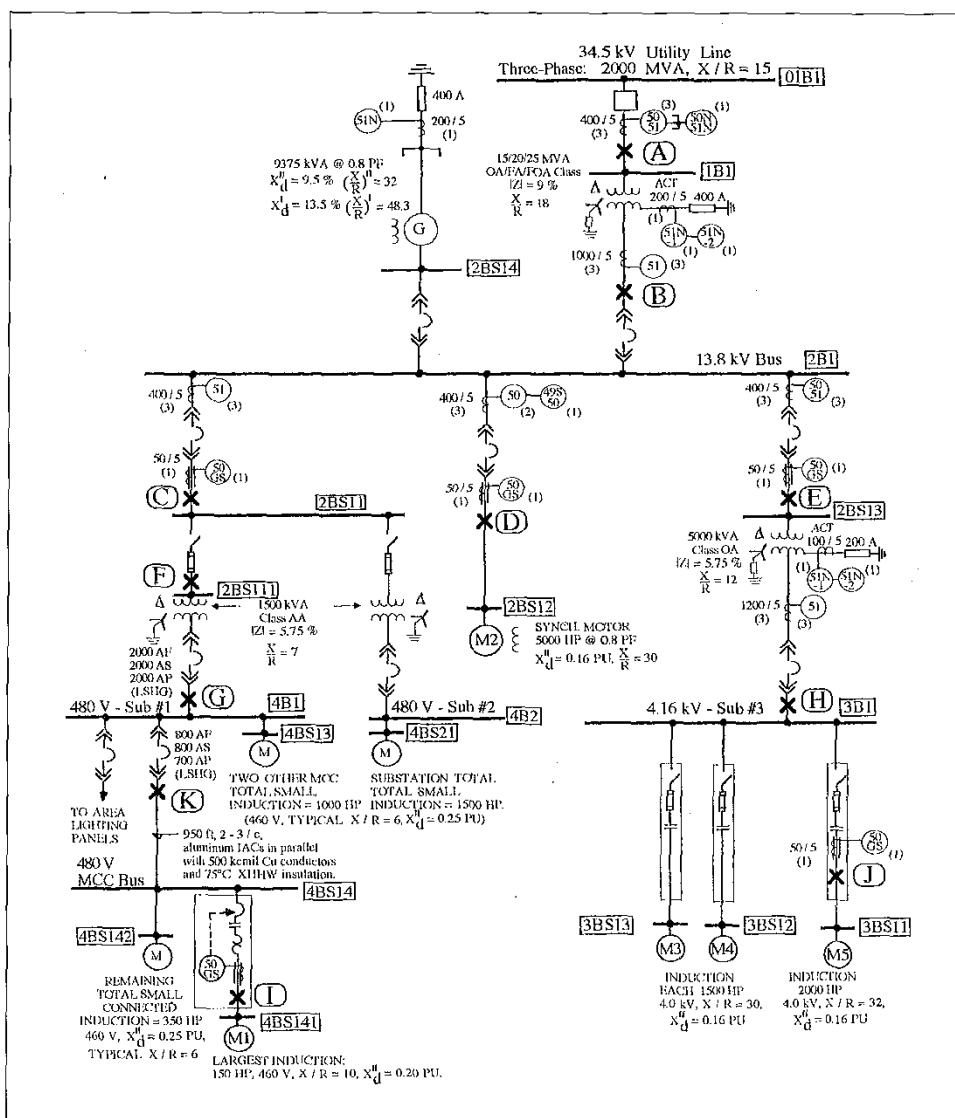


Fig. 4. Labeling of buses for computer-aided short-circuit study.

Table I. Total Symmetrical RMS Kiloamperes for Bolted Three-Phase Faults

Faulted Bus	Prefault Voltage	Maximal First-Cycle	Minimal First-Cycle	Maximal Interrupt	Approx. 30-Cycle
1B1	34.5 kV	34.821	34.775	34.787	34.233
2B1	13.8 kV	13.537	13.091	13.204	9.338
2BS111	13.8 kV	13.493	13.049	13.163	9.316
3B1	4.16 kV	14.258	12.749	12.357	8.683
4B1	480 V	36.871	36.796	36.871	28.099

Table II. Branch Flows of Symmetrical RMS Kiloamperes for Bolted Three-Phase Faults

From - To Buses	Faulted Bus	Maximal First-Cycle	Minimal First-Cycle	Approx. 30-Cycle
2B1 - 1B1	1B1	1.407	1.362	0.820
2BS11 - 2B1	2B1	0.419	0.220	0
2BS12 - 2B1	2B1	1.506	1.505	0
2BS13 - 2B1	2B1	1.053	0.807	0
2BS14 - 2B1	2B1	4.125	4.125	2.902
4B1 - 2BS111	2BS111	0.219	0.219	0
3BS11 - 3B1	3B1	1.967	1.966	0
3BS12 - 3B1	3B1	1.476	0	0
3BS13 - 3B1	3B1	1.475	1.475	0
4BS13 - 4B1	4B1	5.408	5.408	0
4BS14 - 4B1	4B1	2.474	2.474	0
4BS141 - 4BS14	4BS14	1.030	1.030	0

and approximate 30-cycle. The maximal power system configuration includes all possible short-circuit current sources. The minimal power system configuration is defined for this example to exclude (i.e., out-of-service) the 1500 hp M4 induction motor load and the loads of low-voltage substation 2. Finally, the only sources of short-circuit current under the approximate 30-cycle network configuration are the utility tie and in-plant generator; the in-plant generator is represented by its direct-axis transient reactance instead of the subtransient reactance.

Tables I and II provide a summary of the essential results from the print-outs. The program was run for each power system configuration. For each run, one column of the print-out lists for each bus the total symmetrical rms current for a bolted, three-phase fault at that bus. Table I contains only those total symmetrical rms currents that are necessary for the example of Fig. 4. In addition, an important table is provided with the print-out that lists all of the branch flows of first-cycle symmetrical rms current for a bolted, three-phase fault at every bus. Table II contains only those branch flows

of first-cycle symmetrical rms current that are necessary for the example of Fig. 4.

As an example, consider fault location D in Fig. 4. Note that the current transformers for the motor M2 overcurrent relays will not sense the M2 motor backfeed for a three-phase fault at location D. Consequently, the maximal subtransient symmetrical rms current at fault location D, used to select the settings of the motor M2 overcurrent relays, is the following:

$$(I_d'')_{D}^{MAX} = I_{2B1}^{fmax} - I_{2BS12-2B1}^{fmax} \\ = 12.031 \text{ kA @ } 13.8 \text{ kV,}$$

where $I_{2B1}^{fmax} = 13.537 \text{ kA}$ from Table I is the total first-cycle symmetrical rms current for a bolted, three-phase fault at the 13.8 kV main bus 2B1 and $I_{2BS12-2B1}^{fmax} = 1.506 \text{ kA}$ from Table II is the branch flow of first-cycle symmetrical rms current from motor M2 bus 2BS12 to bus 2B1 under a three-phase bolted fault at bus 2B1.

The same rationale is used to derive the following equations to calculate the maximal subtransient sym-

**Table III. Computer-Aided Short-Circuit Study Results
Symmetrical RMS Currents at Fault Locations of Protective Devices**

Protective device fault location	Maximal, subtransient symmetrical rms kA (I''_d) ^{MAX}	Minimal, subtransient symmetrical rms kA (I''_d) ^{MIN}	Maximal, transient symmetrical rms kA (I'_d) ^{MAX}	Minimal, transient symmetrical rms kA (I'_d) ^{MIN}
A	33.414	33.413	N.A.	33.413
B*	6.434	6.434	(6.434)	6.434
C*	13.118	12.871	(12.263)	9.338
D*	12.031	11.586	(11.324)	9.338
E*	12.484	12.284	(11.527)	9.338
F*	13.274	12.830	(12.263)	9.316
G	28.989	28.914	N.A.	28.099
H	9.340	9.307	N.A.	8.683
I	11.662	11.657	N.A.	9.399
J	12.291	10.783	N.A.	8.683
K	34.397	34.322	N.A.	28.099

N.A. - Not available from computer-aided short-circuit study. * - Computer results validated with hand calculations. (*) - Result of hand calculation.

metrical rms currents at the other fault locations. The numerical results are presented in Table III.

$$\begin{aligned}
 (I''_d)_A^{MAX} &= I_{2B1}^{fmax} - I_{2BS11-2B1}^{fmax} \\
 (I''_d)_B^{MAX} &= I_{2B1}^{fmax} - I_{2BS11-2B1}^{fmax} - I_{2BS12-2B1}^{fmax} \\
 &\quad - I_{2BS13-2B1}^{fmax} - I_{2BS14-2B1}^{fmax} \\
 (I''_d)_C^{MAX} &= I_{2B1}^{fmax} - I_{2BS11-2B1}^{fmax} \\
 (I''_d)_E^{MAX} &= I_{2B1}^{fmax} - I_{2BS13-2B1}^{fmax} \\
 (I''_d)_F^{MAX} &= I_{2BS111}^{fmax} - I_{4B1-2BS111}^{fmax} \\
 (I''_d)_G^{MAX} &= I_{4B1}^{fmax} - I_{4BS13-4B1}^{fmax} - I_{4BS14-4B1}^{fmax} \\
 (I''_d)_H^{MAX} &= I_{3B1}^{fmax} - I_{3BS13-3B1}^{fmax} \\
 &\quad - I_{3BS13-3B1}^{fmax} - I_{3BS14-3B1}^{fmax} \\
 (I''_d)_I^{MAX} &= I_{4BS14}^{fmax} - I_{4BS141-4BS14}^{fmax} \\
 (I''_d)_J^{MAX} &= I_{3B1}^{fmax} - I_{3BS11-3B1}^{fmax} \\
 (I''_d)_K^{MAX} &= I_{4B1}^{fmax} - I_{4BS14-4B1}^{fmax}
 \end{aligned}$$

In addition, another column of Table I lists the total interrupting symmetrical rms current for a bolted, three-phase fault at that bus. However, this particular software package does not provide a table in its printout of the branch flows of interrupting symmetrical rms current for a bolted, three-phase fault at every bus. (It only provides a table with branch flows of first-cycle symmetrical rms current.) Without this information, it is not possible to use similar equations to those above to approximate the maximal transient symmetrical

rms currents at the fault locations of the phase-overcurrent-protective devices.

For example, the equation for the maximal transient symmetrical rms current at fault location D (I'_d)_D^{MAX} $\cong$ $I_{2B1}^{fmax} - I_{2BS12-2B1}^{fmax} \cdot I_{2B1}^{fmax} = 13.204$ kA from Table I is the total interrupting symmetrical rms current for a bolted, three-phase fault at 13.8 kV bus 2B1. However, $I_{2BS12-2B1}^{fmax}$, the branch flow of interrupting symmetrical rms current from motor M2 bus 2BS12 to bus 2B1 under a three-phase fault at bus 2B1, is unavailable in the printout of this particular software package. Of course, if the M2 motor backfeed $I_{2BS12-2B1}^{fmax}$ is considered a negligible fraction of I_{2B1}^{fmax} , I_{2B1}^{fmax} is not an overly conservative approximation for (I'_d)_D^{MAX}.

A second run of the software was performed under the minimal power system configuration. The printout of the run provided the same types of results as before, and similar equations to those above were used to calculate the minimal subtransient symmetrical rms currents at the fault locations of the phase-overcurrent-protective devices. The numerical results are presented in Table III.

A third run of the software was performed under the approximate 30-cycle network configuration. Recall that the only sources of short-circuit current under the approximate 30-cycle network configuration are the utility tie and in-plant generator. The in-plant generator is represented by its direct-axis transient reactance instead of its subtransient reactance. Since this version of the software only provides a table in its printout of

branch flows of first-cycle symmetrical rms current, a simple trick was employed to produce a table of branch flows of approximate 30-cycle symmetrical rms current for a bolted, three-phase fault at every bus. The trick was simply to replace the previous input data of the generator's subtransient reactance with the value of its transient reactance (i.e., 13.5%), and rerun the program under the approximate 30-cycle network configuration. The printout of the run provided the same types of results as before (cf. Tables I and II), and similar equations to those above were used to approximate the minimal transient symmetrical rms currents at the fault locations of the phase-overcurrent-protective devices. The results of the calculations are given in Table III.

Hand calculations were performed to validate the results of the computer-aided short-circuit study at fault locations B, C, D, E, and F. Unfortunately, space restrictions disallow their inclusion here.

Summary

There are two types of short-circuit studies of interest to the power engineer. The results of the first type [i.e., maximal first-cycle (momentary) and contact-parting (interrupting) symmetrical rms currents] are used to select the withstand and interrupting capabilities of switchgear, whereas, the results of the second type (i.e., maximal and minimal subtransient and transient symmetrical rms currents) are used to properly select the settings of phase-overcurrent-protective devices. This article presented a comparison of these short-circuit studies for the case of a bolted, three-phase fault. [Other types of faults (e.g., phase-to-phase and phase-to-ground faults) must also be considered when performing a protective-device coordination study.] The results of the comparison were then used to formulate recommendations on how computer-aided results of the first type can be applied to determine the short-circuit currents of the second type.

It is imperative that the engineer clearly understand the solution algorithm of the software. To perform the computations, many of the algorithms employ either reduction of the complex impedance network representation of the power system or else reductions of separate reactance and resistance net-

works. These methods produce similar, but different results. The recommendations of this article are based on the complex impedance network representation of the power system.

This article has discussed the importance of fault location and motor backfeed in determining the short-circuit currents to select the settings of phase-overcurrent-protective devices. It is important to note that accounting for motor backfeed may need to be post-processed by hand from the results of the computer-aided short-circuit study.

Furthermore, a thorough understanding of the possible loading and supply scenarios of the plant is necessary in order to properly define the maximal and minimal power system configurations for the study.

Dedication

Dr. Mercede wishes to dedicate this article to the Blessed Mother of Jesus Christ, under the title of Our Lady of Guadalupe, in gratitude for her intercessory prayers.

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